FOUNDATIONSOF ETHICAL HACKING

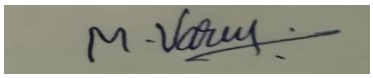
Course Faculty : Dr. M. MAYURANATHAN
Course Name : Ethical Hacking Course
Code : ITA1440
Slot : Slot B
Total Marks : 100

CO1 :

Apply the principle of the foundational concepts of ethical hacking, including types of hackers, ethical principles, legal frameworks, and the role of responsible disclosure in cybersecurity.

Code of Conduct:

I M. Vara Prasad/192324251 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student

Answer all Questions (Each Questions carries 4 marks)

1) Rahul is having coffee with his friend in the college canteen. His friend leaves for the washroom, leaving his mobile phone on the table. Rahul unlocks the phone using his friend's >ingerprint while he is sleeping (previously registered for fun), reads all WhatsApp chats, but does not send any message or delete any data.

A. Section 43 – Unauthorized Access
B. Section 65 – Tampering with Source Code
C. Section 66 – Computer Related Offence (Hacking)
D. Section 66B – Receiving Stolen Computer Resource
E. Section 66C – Identity Theft
F. Section 66D – Cheating by Personation (Online Fraud)
G. Section 66E – Violation of Privacy
H. Section 66F – Cyber Terrorism

2) A Final-year student discovers the default administrator password of the college ERP system. Out of curiosity, he logs in and views all students' internal marks. He neither changes nor

downloads any data.

A. Section 43 – Unauthorized Access
B. Section 65 – Tampering with Source Code
C. Section 66 – Computer Related Offence (Hacking)
D. Section 66B – Receiving Stolen Computer Resource
E. Section 66C – Identity Theft
F. Section 66D – Cheating by Personation (Online Fraud)
G. Section 66E – Violation of Privacy
H. Section 66F – Cyber Terrorism

3) An employee copies confidential customer details from the company database onto a USB drive and later sells the information to a competing company.

A. Section 43 – Unauthorized Access
B. Section 65 – Tampering with Source Code
C. Section 66 – Computer Related Offence (Hacking)
D. Section 66B – Receiving Stolen Computer Resource
E. Section 66C – Identity Theft
F. Section 66D – Cheating by Personation (Online Fraud)
G. Section 66E – Violation of Privacy
H. Section 66F – Cyber Terrorism

4) A software engineer secretly changes the source code of an online banking application so that ₹10 from every transaction is transferred to his own account.

A. Section 43 – Unauthorized Access
B. Section 65 – Tampering with Source Code
C. Section 66 – Computer Related Offence (Hacking)
D. Section 66B – Receiving Stolen Computer Resource
E. Section 66C – Identity Theft
F. Section 66D – Cheating by Personation (Online Fraud)
G. Section 66E – Violation of Privacy
H. Section 66F – Cyber Terrorism

5) A cybercriminal creates a fake SBI login page that looks exactly like the original website. Thousands of users enter their usernames, passwords, and OTPs.

A. Section 43 – Unauthorized Access
B. Section 65 – Tampering with Source Code
C. Section 66 – Computer Related Offence (Hacking)
D. Section 66B – Receiving Stolen Computer Resource
E. Section 66C – Identity Theft
F. Section 66D – Cheating by Personation (Online Fraud)
G. Section 66E – Violation of Privacy
H. Section 66F – Cyber Terrorism

6) A student installs a keylogger on the department computer to capture the usernames and passwords of faculty members.
A. Section 43 – Unauthorized Access
B. Section 65 – Tampering with Source Code

C. Section 66 – Computer Related Offence (Hacking)
D. Section 66B – Receiving Stolen Computer Resource
E. Section 66C – Identity Theft
F. Section 66D – Cheating by Personation (Online Fraud)
G. Section 66E – Violation of Privacy
H. Section 66F – Cyber Terrorism

7) A person knowingly purchases a stolen of>ice laptop containing con>idential company information because it is available at a very low price.

A. Section 43 – Unauthorized Access
B. Section 65 – Tampering with Source Code
C. Section 66 – Computer Related Offence (Hacking)
D. Section 66B – Receiving Stolen Computer Resource
E. Section 66C – Identity Theft
F. Section 66D – Cheating by Personation (Online Fraud)
G. Section 66E – Violation of Privacy
H. Section 66F – Cyber Terrorism

8) A student secretly records another student inside the hostel bathroom and shares the video in a Telegram group.

A. Section 43 – Unauthorized Access
B. Section 65 – Tampering with Source Code
C. Section 66 – Computer Related Offence (Hacking)
D. Section 66B – Receiving Stolen Computer Resource
E. Section 66C – Identity Theft
F. Section 66D – Cheating by Personation (Online Fraud)
G. Section 66E – Violation of Privacy
H. Section 66F – Cyber Terrorism

9) A hacker gains access to a university server, downloads examination question papers, and deletes the original >iles before leaving.

A. Section 43 – Unauthorized Access
B. Section 65 – Tampering with Source Code
C. Section 66 – Computer Related Offence (Hacking)
D. Section 66B – Receiving Stolen Computer Resource
E. Section 66C – Identity Theft
F. Section 66D – Cheating by Personation (Online Fraud)
G. Section 66E – Violation of Privacy
H. Section 66F – Cyber Terrorism

10) An employee leaks con>idential patient medical records to a news agency after completing an authorized security audit.

A. Section 43 – Unauthorized Access
B. Section 65 – Tampering with Source Code
C. Section 66 – Computer Related Offence (Hacking)
D. Section 66B – Receiving Stolen Computer Resource
E. Section 66C – Identity Theft

F. Section 66D – Cheating by Personation (Online Fraud)
G. Section 66E – Violation of Privacy
H. Section 66F – Cyber Terrorism

11) A hacker launches ransomware against a private hospital, encrypting all patient records and demanding ₹50 lakh to restore access.

A. Section 43 – Unauthorized Access
B. Section 65 – Tampering with Source Code
C. Section 66 – Computer Related Offence (Hacking)
D. Section 66B – Receiving Stolen Computer Resource
E. Section 66C – Identity Theft
F. Section 66D – Cheating by Personation (Online Fraud)
G. Section 66E – Violation of Privacy
H. Section 66F – Cyber Terrorism

12) A fraudster calls senior citizens pretending to be a bank manager and convinces them to share OTPs, stealing money from their accounts.

A. Section 43 – Unauthorized Access
B. Section 65 – Tampering with Source Code
C. Section 66 – Computer Related Offence (Hacking)
D. Section 66B – Receiving Stolen Computer Resource
E. Section 66C – Identity Theft
F. Section 66D – Cheating by Personation (Online Fraud)
G. Section 66E – Violation of Privacy
H. Section 66F – Cyber Terrorism

13) A college student modi>ies the attendance software source code so that his attendance automatically becomes 100%.

A. Section 43 – Unauthorized Access
B. Section 65 – Tampering with Source Code
C. Section 66 – Computer Related Offence (Hacking)
D. Section 66B – Receiving Stolen Computer Resource
E. Section 66C – Identity Theft
F. Section 66D – Cheating by Personation (Online Fraud)
G. Section 66E – Violation of Privacy
H. Section 66F – Cyber Terrorism

14) A hacker replaces the homepage of a government website with political slogans and threatening messages.

A. Section 43 – Unauthorized Access
B. Section 65 – Tampering with Source Code
C. Section 66 – Computer Related Offence (Hacking)
D. Section 66B – Receiving Stolen Computer Resource
E. Section 66C – Identity Theft
F. Section 66D – Cheating by Personation (Online Fraud)
G. Section 66E – Violation of Privacy
H. Section 66F – Cyber Terrorism

15) A software tester copies the unreleased source code of his company and uploads it to GitHub after resigning.

A. Section 43 – Unauthorized Access
B. Section 65 – Tampering with Source Code
C. Section 66 – Computer Related Offence (Hacking)
D. Section 66B – Receiving Stolen Computer Resource
E. Section 66C – Identity Theft
F. Section 66D – Cheating by Personation (Online Fraud)
G. Section 66E – Violation of Privacy
H. Section 66F – Cyber Terrorism

16) A person uses another individual's Aadhaar number and mobile OTP to open a fake bank account.

A. Section 43 – Unauthorized Access
B. Section 65 – Tampering with Source Code
C. Section 66 – Computer Related Offence (Hacking)
D. Section 66B – Receiving Stolen Computer Resource
E. Section 66C – Identity Theft
F. Section 66D – Cheating by Personation (Online Fraud)
G. Section 66E – Violation of Privacy
H. Section 66F – Cyber Terrorism

17) A student accesses the college Wi-Fi controller without authorization and disconnects internet access for the entire campus.

A. Section 43 – Unauthorized Access
B. Section 65 – Tampering with Source Code
C. Section 66 – Computer Related Offence (Hacking)
D. Section 66B – Receiving Stolen Computer Resource
E. Section 66C – Identity Theft
F. Section 66D – Cheating by Personation (Online Fraud)
G. Section 66E – Violation of Privacy
H. Section 66F – Cyber Terrorism

18) A hacker attacks the control system of an electricity distribution station to create a citywide power outage.

A. Section 43 – Unauthorized Access
B. Section 65 – Tampering with Source Code
C. Section 66 – Computer Related Offence (Hacking)
D. Section 66B – Receiving Stolen Computer Resource
E. Section 66C – Identity Theft
F. Section 66D – Cheating by Personation (Online Fraud)
G. Section 66E – Violation of Privacy
H. Section 66F – Cyber Terrorism

19) A company gives an ethical hacker permission to test only its website. Instead, he also scans the company's internal HR server without approval.

A. Section 43 – Unauthorized Access
B. Section 65 – Tampering with Source Code
C. Section 66 – Computer Related Offence (Hacking)
D. Section 66B – Receiving Stolen Computer Resource
E. Section 66C – Identity Theft
F. Section 66D – Cheating by Personation (Online Fraud)
G. Section 66E – Violation of Privacy
H. Section 66F – Cyber Terrorism

20) A student discovers his teacher's Gmail password written on a sticky note and logs in to view con>idential question papers.

A. Section 43 – Unauthorized Access
B. Section 65 – Tampering with Source Code
C. Section 66 – Computer Related Offence (Hacking)
D. Section 66B – Receiving Stolen Computer Resource
E. Section 66C – Identity Theft
F. Section 66D – Cheating by Personation (Online Fraud)
G. Section 66E – Violation of Privacy
H. Section 66F – Cyber Terrorism

21) A person edits another person's private photographs using AI and shares the manipulated images online to humiliate them.

A. Section 43 – Unauthorized Access
B. Section 65 – Tampering with Source Code
C. Section 66 – Computer Related Offence (Hacking)
D. Section 66B – Receiving Stolen Computer Resource
E. Section 66C – Identity Theft
F. Section 66D – Cheating by Personation (Online Fraud)
G. Section 66E – Violation of Privacy
H. Section 66F – Cyber Terrorism

22) A hacker creates fake customer-care numbers on Google Search and tricks users into installing remote-access software.

A. Section 43 – Unauthorized Access
B. Section 65 – Tampering with Source Code
C. Section 66 – Computer Related Offence (Hacking)
D. Section 66B – Receiving Stolen Computer Resource
E. Section 66C – Identity Theft
F. Section 66D – Cheating by Personation (Online Fraud)
G. Section 66E – Violation of Privacy
H. Section 66F – Cyber Terrorism

23) A government employee intentionally deletes source code used in a public service portal before retirement.

A. Section 43 – Unauthorized Access
B. Section 65 – Tampering with Source Code
C. Section 66 – Computer Related Offence (Hacking)

D. Section 66B – Receiving Stolen Computer Resource
E. Section 66C – Identity Theft
F. Section 66D – Cheating by Personation (Online Fraud)
G. Section 66E – Violation of Privacy
H. Section 66F – Cyber Terrorism

24) A hacker accesses a military communication server and attempts to disrupt defence operations.

A. Section 43 – Unauthorized Access
B. Section 65 – Tampering with Source Code
C. Section 66 – Computer Related Offence (Hacking)
D. Section 66B – Receiving Stolen Computer Resource
E. Section 66C – Identity Theft
F. Section 66D – Cheating by Personation (Online Fraud)
G. Section 66E – Violation of Privacy
H. Section 66F – Cyber Terrorism

25) A college student writes a worm that automatically spreads to every computer in the laboratory, slowing down all systems.

A. Section 43 – Unauthorized Access
B. Section 65 – Tampering with Source Code
C. Section 66 – Computer Related Offence (Hacking)
D. Section 66B – Receiving Stolen Computer Resource
E. Section 66C – Identity Theft
F. Section 66D – Cheating by Personation (Online Fraud)
G. Section 66E – Violation of Privacy
H. Section 66F – Cyber Terrorism

RUBRICS

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (ProWicient)	Level 2 (Developing)	Level 1 (Beginning)
IdentiWication of Cybersecurity Offences	25%	Correctly identi>ies all relevant cyber offences and applicable legal provisions for the given scenarios.	Correctly identi>ies most offences with minor errors.	Identi>ies some offences but confuses applicable provisions in several cases.	Fails to identify major cyber offences or applicable provisions.
Application of Legal Frameworks	25%	Accurately applies relevant legal provisions to each scenario with clear understanding ofthe underlying offence.	Applies legal provisions correctly to most scenarios with minor inaccuracies.	Applies some provisions correctly but provides limited understanding of their applicability.	Unable to appropriate ly apply legal provisions to the scenarios.

Ethical and Security Interpretation	30%	Demonstrates excellent understanding of unauthorized access, privacy, integrity, identity theft, cybercrime, and ethical responsibilities through accurate scenario interpretation.	Correctly interprets most ethical and security issues with minor omissions.	Identi>ies basic ethical/security issues but provides limited interpretation.	Demonstrates little understanding of the ethical or security implications.
Accuracy and Decision Making	20%	Provides consistently accurate decisions and demonstrates strong reasoning when distinguishing between different cyber offences.	Provides mostly accurate decisions with minor mistakes.	Provides partially correct decisions with several inconsistencies.	Responses are largely incorrect, incomplete, or unrelated to the scenarios.

Answers with reasons and ethical points

1. Rahul unlocks his friend's phone and reads WhatsApp chats without permission.

Correct answer: G. Section 66E – Violation of Privacy

Reason: Rahul accesses his friend's private WhatsApp conversations without consent. This is an invasion of privacy, even though he does not send messages or delete data.

Ethical point: Having physical access to someone's phone does not mean having permission to read their private messages.

2. A student uses the default administrator password to view students' internal marks without permission.

Correct answer: A. Section 43 – Unauthorized Access

Reason: The student accesses the college ERP system without authorization. Viewing confidential marks is unauthorized even if no data is changed or downloaded.

Ethical point: Discovering a weak password does not give someone permission to use it.

3. An employee copies confidential customer details to a USB drive and sells them to a competitor.

Correct answer: A. Section 43 – Unauthorized Access

Reason: Section 43 covers unauthorized acts involving computer data, including copying or extracting information without permission. Selling the information also constitutes a serious breach of confidentiality and may attract other applicable laws.

Ethical point: Employees must protect customer information and must not misuse their access for personal gain.

4. A software engineer secretly modifies banking application code to transfer ₹10 from every transaction to his account.

Correct answer: B. Section 65 – Tampering with Source Code

Reason: The engineer secretly alters software code to manipulate financial transactions. This involves unauthorized code modification and fraudulent activity. Section 65 specifically concerns tampering with computer source documents, so its precise applicability depends on the nature of the code and conduct; other provisions may also apply.

Ethical point: Software engineers must maintain the integrity and security of financial systems.

5. A cybercriminal creates a fake SBI login page to steal usernames, passwords, and OTPs.

Correct answer: F. Section 66D – Cheating by Personation

Reason: The criminal impersonates the genuine SBI website to deceive users into revealing confidential banking credentials. This is a phishing attack involving online impersonation and fraud.

Ethical point: Websites must not be impersonated to deceive users or steal their financial information.

6. A student installs a keylogger to capture faculty members' usernames and passwords.

Correct answer: E. Section 66C – Identity Theft

Reason: A keylogger secretly records keystrokes to capture login credentials. Using another person's credentials dishonestly can constitute identity theft under Section 66C. Installing the keylogger may also violate Section 43 and, depending on the circumstances, Section 66.

Ethical point: Login credentials must remain confidential and must never be collected without authorization.

7. A person knowingly purchases a stolen office laptop containing confidential company information.

Correct answer: D. Section 66B – Receiving Stolen Computer Resource

Reason: Section 66B addresses dishonestly receiving or retaining a stolen computer resource or communication device, knowing or having reason to believe it is stolen. The scenario specifically states that the person knowingly purchases the stolen laptop.

Ethical point: People should verify the ownership and legitimate source of electronic devices before purchasing them.

8. A student secretly records another student in a hostel bathroom and shares the video on Telegram.

Correct answer: G. Section 66E – Violation of Privacy

Reason: The student secretly records someone in a private setting and distributes the video without consent. This is a serious invasion of privacy. Section 66E specifically concerns capturing, publishing, or transmitting images of a person's private area in circumstances violating privacy; its application depends on the actual content and circumstances.

Ethical point: Recording or sharing intimate images without consent violates privacy and can cause serious emotional and reputational harm.

9. A hacker downloads examination question papers from a university server and deletes the original files.

Correct answer: C. Section 66 – Computer Related Offence

Reason: The hacker accesses the server without authorization, extracts confidential examination papers, and deletes files. These dishonest acts may attract Section 66 when its requirements are met, along with Section 43 for unauthorized access, copying, and damage to data.

Ethical point: Unauthorized access and destruction of academic records compromise data confidentiality, integrity, and availability.

10. An employee leaks confidential patient medical records to a news agency after completing an authorized security audit.

Correct answer: G. Section 66E – Violation of Privacy

Reason: The employee discloses confidential patient information without authorization, violating the patients' privacy and confidentiality. However, Section 66E specifically concerns images of private areas, so it may not directly apply to medical records alone. Depending on the circumstances, other provisions concerning unauthorized disclosure or breach of confidentiality may be more appropriate.

Ethical point: Authorization to conduct a security audit does not grant permission to disclose confidential patient records. Security professionals must maintain confidentiality and report findings through authorized channels.

11. A hacker uses ransomware against a hospital and demands ₹50 lakh.

Answer: C. Section 66 – Computer Related Offence

Reason: The hacker encrypts patient records and demands payment. This is malicious computer activity that may involve unauthorized data damage and disruption under Section 43 and, where applicable, Section 66.

Ethical point: Attacking hospital systems can endanger patients and disrupt essential medical services.

12. A fraudster impersonates a bank manager to steal OTPs from senior citizens.

Answer: F. Section 66D – Cheating by Personation

Reason: The fraudster pretends to be a bank manager to deceive victims into revealing OTPs and transferring money.

Ethical point: Never impersonate trusted professionals to deceive people or steal their money.

13. A student modifies attendance software to show 100% attendance.

Answer: B. Section 65 – Tampering with Source Code

Reason: The student alters software to falsify attendance records. Section 65 may apply if computer source documents are tampered with.

Ethical point: Academic records must remain accurate and must not be manipulated for personal benefit.

14. A hacker replaces a government website homepage with political slogans and threats.

Answer: C. Section 66 – Computer Related Offence

Reason: Unauthorized website defacement may involve unauthorized access or data alteration under Section 43, with Section 66 potentially applying when its requirements are met. Section 66F applies only if the specific requirements for cyberterrorism are satisfied.

Ethical point: Government websites must not be accessed or altered without authorization.

15. A software tester uploads unreleased company source code to GitHub after resigning.

Answer: A. Section 43 – Unauthorized Access

Reason: Copying or extracting confidential source code without authorization may fall under Section 43. Depending on the circumstances, confidentiality, intellectual-property, and other legal provisions may apply.

Ethical point: Resigning from a company does not give an employee permission to disclose its confidential code.

16. A person uses another individual's Aadhaar number and OTP to open a fake bank account.

Answer: E. Section 66C – Identity Theft

Reason: The person misuses another individual's identity information and electronic credentials. Section 66D may also apply if the conduct involves cheating by personation.

Ethical point: Personal identity information and OTPs must never be misused.

17. A student accesses the college Wi-Fi controller and disconnects internet access campus-wide.

Answer: C. Section 66 – Computer Related Offence

Reason: Unauthorized access and disruption of network services may violate Section 43. Section 66 may apply when the relevant conduct is dishonest or fraudulent.

Ethical point: Network administrators' systems must only be accessed by authorized personnel.

18. A hacker attacks an electricity distribution system to cause a citywide blackout.

Answer: H. Section 66F – Cyber Terrorism

Reason: The attack targets critical infrastructure and could threaten public safety or national security. Section 66F requires specific statutory conditions; not every infrastructure attack automatically qualifies.

Ethical point: Critical infrastructure must be protected because attacks can endanger lives and disrupt essential services.

19. An ethical hacker tests a website but scans the internal HR server without permission.

Answer: A. Section 43 – Unauthorized Access

Reason: The company authorized testing only the website. Scanning another system exceeds the agreed scope and may constitute unauthorized access or related unauthorized activity, depending on what was done.

Ethical point: Ethical hackers must follow the approved scope of work and obtain permission before testing additional systems.

20. A student uses a teacher's Gmail password to access confidential examination papers.

Answer: E. Section 66C – Identity Theft

Reason: The student dishonestly uses another person's password to access an account. Section 43 may also apply to the unauthorized access.

Ethical point: Finding or knowing someone's password does not grant permission to use their account.

21. A person uses AI to manipulate private photographs and shares them to humiliate someone.

Answer: G. Section 66E – Violation of Privacy

Reason: Manipulating and distributing private images without consent can violate privacy and may attract other applicable laws. Section 66E specifically concerns certain images of private areas, so it is not automatically applicable to every manipulated photograph.

Ethical point: AI must not be used to humiliate, impersonate, or harm another person.

22. A hacker creates fake customer-care numbers and tricks users into installing remote-access software.

Answer: F. Section 66D – Cheating by Personation

Reason: The hacker impersonates customer-care representatives and deceives users into installing software that may enable fraud or unauthorized access.

Ethical point: Customer-care services must not be impersonated to deceive users or steal information.

23. A government employee intentionally deletes source code from a public-service portal.

Answer: B. Section 65 – Tampering with Source Code

Reason: The employee deliberately deletes source code. Section 65 may apply if the conduct involves tampering with computer source documents; Section 43 may also be relevant to unauthorized damage or deletion.

Ethical point: Government employees must protect public-service systems and preserve important digital records.

24. A hacker accesses a military communication server to disrupt defence operations.

Answer: H. Section 66F – Cyber Terrorism

Reason: The target is a military communication server, and the stated objective is to disrupt defence operations. This may qualify as cyberterrorism if the statutory requirements are satisfied.

Ethical point: Military systems and defence communications must be protected against unauthorized access and disruption.

25. A student creates a worm that spreads across laboratory computers and slows them down.

Answer: C. Section 66 – Computer Related Offence

Reason: The worm spreads maliciously and disrupts computer systems. Section 43 may cover unauthorized damage or disruption, while Section 66 may apply if the relevant acts are committed dishonestly or fraudulently.

Ethical point: Malware must not be developed or deployed against systems without proper authorization.